Tianxin Tao

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TECHNICAL SKILLS

Programming Language	Software	Electrical Equipment and Device	Mechanical Equipment and Machine
C, C++, C#	Git	Raspberry Pi	Lathes
Python	Pybullet	Microcomputer	Water-jet Cutter
MATLAB	CLion	(Altera DE0-CV)	Hydraulic System
Java	WebStorm	Microprocessor(TI MSP	3D Printer
SQL	Three.js	Series)	
JavaScript	Gazebo	Arduino	
VHDL	MATLAB (Simulink)		
HTML, CSS	Jupyter Notebook		

EDUCATION

M.Sc. in Computer Science — **To begin in September, 2019**
The University of British Columbia, Vancouver

BASc in Mechanical Engineering with minor in computer science — **September, 2014 – April, 2019**
The University of British Columbia, Vancouver
GPA: 4.30/4.33

PERSONAL PROJECTS

SNC-Lavalin project – PIMS BC Data Science Workshop — **June, 2018**

- Interpolated the ship path by cubic spline
- Implemented the KNN (k-nearest neighbors) algorithm to reconstruct the sparse history ship path using the dense current ship path
- Preprocessed the dataset to fit problem into a recommendation problem by filtering, grouping and augmenting the dataset

Development of an ICT Platform for Water Quality Monitoring (IC-IMPACTS) — **May, 2017 – September, 2017**

- Implemented boustrophedon decomposition path planning algorithm
- Visualized the state information of an AR.Drone from ROS in rviz and displayed the path and sampling points of the AR.Drone using python
- Simulated the motion of an AR.Drone using python in Gazebo
- Controlled the motor speed of an automation boat using C through a Raspberry Pi

ACADEMIC PROJECTS

Manual Wheelchair User Intention Detection (UBC CARIS Lab) — **September, 2018 - present**

- Collected motion data of the wheelchair from iPhone sensors by MATLAB
- Derived angular acceleration from angular velocity to augment the dataset using 5-point numerical differentiation scheme
- Classified user's intention using Bayesian based approach

UCI Default of Credit Card Clients — **April, 2018**

- Cleaned up the dataset and preprocessed the features by transforming non-numerical variables, feature scaling and change of basis

- Tested performance of different classifiers such as logistic regression, random forest and MLP
- Tuned the hyperparameters using grid search

Online Shopping **January, 2018 – April, 2018**

- Designed and built the schema of the store data in an efficient way using Oracle SQL queries
- Developed a friendly UI for users to interact by Java Swing
- Implemented a variety of queries to retrieve data from the database to be presented to the user

Buses Are Us - Android Application Project **August, 2017**

- Developed an Android App that maps the location of stops and bus routes on the Greater Vancouver Transit system
- Parsed through Translink arrival and departure data to display schedule of the bus routes
- Implemented the JUnit test framework to test the functionality of each component

Regenerator Miniature Prototype **January, 2016**

- Generated concepts to stop and restart the vehicle with the energy recovered from braking
- Evaluated the concepts by estimating and calculating the key parameters of different concepts, for example, the spring constant, weight of the vehicle, and acceleration
- Built the regenerator miniature prototype and conducted tests

WORK EXPERIENCE

UBC Department of Forest Resources Management **May, 2018 - August, 2018**

Developer

- Designed the user interface of the website and implemented the frontend using ReactJS
- Implemented user sign up, log in functionality using Firebase
- Designed the schema of the database and set up the database rules to restrict user access to the database

VOLUNTEER WORK EXPERIENCE

UBC Industrial and Automation Laboratory **May, 2017 – September, 2017**

Research Volunteer

- Implemented simulation and physical experiments to exam the creative algorithms that the supervisor proposed to acquire the water-quality model of a lake with a relatively smaller number of sample points

UBC Applied Science Open House **November, 2015**

- Introduced the curriculum to prospective engineering students
- Exhibited Maglev (project built in Mech 220) to the public

PUBLICATIONS

Towards the Development of a Learning-Based Intention Detection Framework for Pushrim-Activated Power-Assisted Wheelchairs

Mahsa Khalili*, Tianxin Tao, Ruolan Ye, Shuyong Xie, Huancheng Yang, H.F. Machiel Van der Loos, Jaimie Borisoff

2019 IEEE-RAS-EMBS International Conference on Rehabilitation Robotics (ICORR) (To appear)

AWARDS AND SCHOLARSHIPS

Faculty of Applied Science International Student Scholarship **December, 2018**

Offered to continuing international students in the Faculty of Applied Science who demonstrate strong academic achievement, engagement in the Faculty, and the potential to make a scholarly contribution within their chosen field of study.

Elizabeth and Leslie GOULD Scholarship in Engineering **November, 2018**
Three scholarships offered to engineering students in the Faculty of Applied Science which are made on the recommendation of the Faculty

Undergraduate Student Academic Achievement Award (Mechanical Engineering Department) **October, 2018**
October, 2017
Offered to students with top academic performance

The Trek Excellence Scholarship **September, 2018**
September, 2015
Offered to students in the top 5% of their undergraduate year